WORKSHEET

Variance and standard deviation

1 An eight-sided die is rolled and random variable X = the score. Find the:

a expected value of X

b variance of X

c standard deviation of X.

2 Find the variance and standard deviation of the following probability distributions.

a

x	1	2	3	4
$p(x)$	0.1	0.2	0.3	0.4

b

x	0	1	2	3
$p(x)$	0.15	0.2	0.35	0.3

c

x	600	800	1000
$p(x)$	0.2	0.3	0.5

d

x	56	63	71	100
$P(X = x)$	0.16	0.32	0.5	0.02

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3 The values of random variable Y are: 7, 5, 5, 7, 9, 8, 3, 6, 8, 6, 7, 5, 7, 6, 6, 10, 6, 8, 7, 9.

a Draw a graph of the probability distribution for Y.

b Find $E(Y)$.

c Find $Var(Y)$.

d Find $SD(Y)$.

4 Find the mean, variance and standard deviation for the following sets of test scores:

Class A: 74, 80, 93, 85, 81, 79, 82, 88, 75, 79

Class B: 80, 52, 98, 76, 91, 65, 76, 42, 55, 89

5 Two six-sided dice are rolled and Z = the difference between the two scores.

a Construct a probability distribution for Z.

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b Calculate $E(Z)$.

c Calculate $Var(Z)$.

6 A coin is tossed four times and T = the number of successive tails.

a Construct a probability distribution for T.

b Calculate $E(T)$.

c Calculate $Var(T)$.

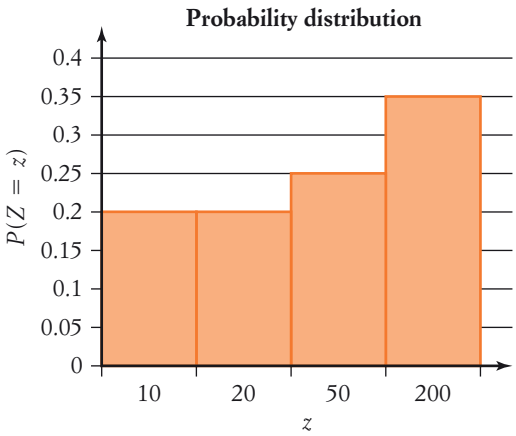
7 The heights (cm) of students were recorded as follows:

155, 162, 150, 173, 164, 177, 146, 159, 143, 168, 184, 163

Find the mean, variance and standard variation of student heights.

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8 Find the expected value, variance and standard deviation of the random variable, Z, with distribution shown in the graph below.



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Answers

1 **a** 4.5

b 5.25

c ≈ 2.29

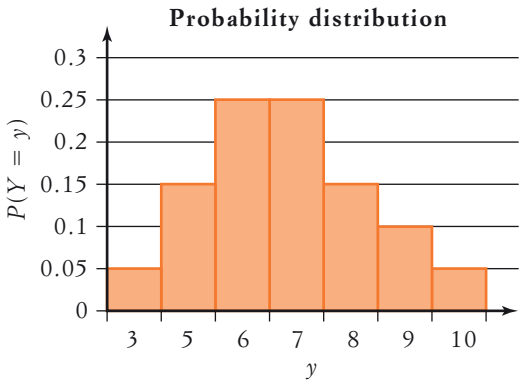
2 **a** $Var(X) = 1$ $SD(X) = 1$

b $Var(X) = 1.06$ $SD(X) \approx 1.03$

c $Var(X) = 24\,400$ $SD(X) \approx 156.21$

d $Var(X) \approx 54.12$ $SD(X) \approx 7.36$

3 **a**



b 6.75

c ≈ 2.59

d ≈ 1.61

4 Class A: mean = 81.6 $Var = 30.04$ $SD \approx 5.48$

Class B: mean = 72.4 $Var = 305.84$ $SD \approx 17.49$

5 **a**

z	0	1	2	3	4	5
$p(z)$	$\frac{1}{6}$	$\frac{5}{18}$	$\frac{2}{9}$	$\frac{1}{6}$	$\frac{1}{9}$	$\frac{1}{18}$

b ≈ 1.94

c ≈ 2.05

6 **a**

t	0	1	2	3	4
$p(t)$	$\frac{1}{16}$	$\frac{7}{16}$	$\frac{5}{16}$	$\frac{1}{8}$	$\frac{1}{16}$

b ≈ 1.69

c ≈ 0.96

7 mean = 162 cm Variance ≈ 140.83 cm^2 $SD \approx 11.87$ cm

8 $E(Z) = 88.5$ $Var(Z) = 6892.75$ $SD(Z) \approx 83.02$

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